

The Red/Black Game II:

Solving the Belief-Reset Abstraction Exactly, and Measuring the Distance to the Real Game

(Setup, Algorithm, Metrology, Two Solves, and What They Cannot Certify)

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Abstract

Paper I proved that heads-up Red/Black has 2.4×10^{33} perfect-recall infosets, so the real game G cannot be engaged by tabulation, and that its public class summarizes knowledge exactly but belief not at all. This paper solves — exactly, twice — the game those facts make natural: G' , Red/Black with beliefs reset to the class prior at every round boundary. It is written for a reader who has never seen counterfactual regret minimization (Section 3 builds CFR+ from the definition of regret), and its contributions come in matched pairs of construction and measurement. We give the decomposition over the 5,721-class layered DAG, the row-level information structure, and the vector-form solver; a first full solve at $\text{tol} = 10^{-3}$ whose own convergence meter reported uniform success while seat-mirror antisymmetry — which the meter cannot see — was violated by up to 0.114; and a second-generation solver (ordering-collapsed private states, seat-mirror reflection, a compiled kernel) that re-solves the entire game on a laptop in two hours at $\text{tol} = 10^{-5}..10^{-4}$ with mirrors exact by construction. The value of G' is $+0.0245$: **51.22%** to the opening seat. The paper’s second half measures the distance from G' to G with validated instruments, and the distance is large: the reach-weighted total-variation between the class prior and the true public posterior is 0.842; a best-response probe certifies real-game exploitability $\varepsilon_G \geq 0.598$ per game for the solved profile while its G' -exploitability meters read 10^{-4} ; and the G' equilibrium provably plays a real-game-dominated action (the suicide flip) on path with probability 0.97, because the belief reset launders the information cost of revealing one’s own cards. Solving the abstraction, exactly, is demonstrably not solving the game — and the demonstration is this laboratory’s point: every claim above is measured, not estimated. Closing the gap is Paper III’s subject.

1 Introduction

Paper I described heads-up Red/Black exactly: a deterministic, acyclic transition system whose public projection at a round boundary is two integers per seat plus the identity of the player owing a discard. That paper deliberately said nothing about how to play. This one does — and then measures, with validated instruments, exactly how far “playing well in the abstraction” sits from playing well in the game.

The target of the first half is an ε -equilibrium of the belief-reset abstraction G' , with ε measured rather than assumed. The method is counterfactual regret minimization, in its CFR+ variant. The interesting parts are not the algorithm, which is standard, but the decomposition it runs on, the metrology attached to it, and — the second half — what the finished, exact solve of G' turned out *not* to mean. Three of this paper’s results are negative, all three are quantitative, and we consider them the most exportable things in it: convergence meters certify the abstraction rather than the game (Section 9.2); sibling-profile head-to-head play is not a solve meter (Incident, Section 9.6); and an exactly converged equilibrium of G' plays a real-game-dominated action on path with probability 0.97 (Theorem 9.1).

Why decompose at all. By Paper I the real game has 5.3×10^{34} public decision histories and 2.4×10^{33} perfect-recall infosets: there is no regret table to hold. Even the class-collapsed game is large — 6,051 public classes and, layer by layer, over 3×10^8 (decision node $\times$ private state) rows — but it is finite in the useful sense. We exploit the fact — Paper I, Theorem 3.7 — that the class space is a *layered directed acyclic graph*: every round removes exactly one card, so classes at layer T' can only lead to classes at layer $T' - 1$. That licenses backward induction. We solve the smallest layer first, then use its answers as terminal payoffs for the layer above, and so on up to the root.

What this buys and what it costs. It buys tractability: each subgame is small enough to solve to high precision, and the layers are embarrassingly parallel within themselves. It costs exactness. A subgame needs to know the distribution over its own starting states, and that distribution is re-derived from the public class rather than carried down from above. Paper I proved (its support-completeness theorem, Thm. 11.1 there, and the surrounding discussion) that the public class determines the *support* of that distribution exactly but not its *shape*. So the decomposition is an approximation whose size must be measured, not argued. Section 7 states this precisely and Section 8 says how it is measured.

Scope. Heads-up ($N = 2$) throughout, hand size $H = 5$, notation continuous with Paper I.

Road map. Section 2 recaps what Paper I established. Section 3 develops CFR and CFR+ from scratch. Section 4 specifies the information structure and defines a row. Section 5 gives the decomposition. Section 6 gives the vector-form implementation and the second-generation solver (ordering collapse, reflection, the kernel). Section 7 states what is approximated, Section 8 defines the meters — including the two that speak about the real game — and Section 9 reports both solves, the selection experiments, and the measured distance to G . Appendix A documents an earlier boundary choice that failed, and why. Throughout, boxed *incidents* record measured mistakes — each one is a general lesson about decomposition solving that cost this project real time, preserved in citable form.

2 What Paper I Established

Four results carry over and are used without reproof.

The game is a deterministic acyclic transition system. All chance is in the deal; every transition is a function of state and action. Total cards strictly decrease, so no state recurs and every play terminates in between 5 and 9 rounds.

Two clocks. The rules order a round *reorder* $\rightarrow$ *bid* $\rightarrow$ *procure* $\rightarrow$ *discard*, so the discard is last. We cut the game at *boundaries* — procurement resolutions, where the loser is known but has not yet discarded — so the segment between two boundaries reads *discard* $\rightarrow$ *reorder* $\rightarrow$ *bid* $\rightarrow$ *procure*, with the discard first. A boundary's hand sizes n_i are therefore *pre-discard*, while the round that follows is played with $n'_i = n_i - [i = \ell]$ and $T' = n'_0 + n'_1$. The layer index is T' .

The public class. At a boundary the public state is (ℓ, s_0, s_1) with $s_i = (n_i, r_{\text{lb}}^i, b_{\text{lb}}^i)$: who owes a discard, each seat's pre-discard hand size, and the certainty bounds. There are

$2K(H)^2 + 1 = 6,051$ such classes at $H = 5$, of which 330 are terminal and 5,485 are reachable.

Knowledge is not belief. The bounds pin the *support* of the public posterior exactly — sound and tight — but say nothing about its shape. Two histories reaching the same class can leave an observer with materially different beliefs.

The game itself is beyond tabulation. 5.3×10^{34} public decision histories; 2.4×10^{33} perfect-recall info sets, counted exactly. Whatever engages G must summarize.

Reachable beliefs factorize. Under any profile the boundary posterior is the class prior times two per-seat tilts (Paper I, Thm. 11.3), with opponent-model-free own-side conditionals. This paper does *not* use that structure — resetting to the prior is precisely discarding the tilts — which is what Section 9.6 prices. Paper III uses it.

The knowledge-is-not-belief point constrains everything in this paper; the factorization point is the door it leaves open.

3 Regret, Counterfactual Regret, and CFR+

This section assumes no prior exposure. It builds the algorithm in five steps and then states, without proving, what it guarantees.

3.1 Regret in a one-shot game

Suppose you play the same one-shot game repeatedly. On iteration t you choose an action a from a finite set A , and afterwards you learn what every action *would* have paid, $u^t(a)$.

Definition 3.1 (Regret). After T iterations, your *cumulative regret* for action a is

$$R^T(a) = \sum_{t=1}^T \left(u^t(a) - u^t(\sigma^t) \right),$$

the total extra payoff you would have collected by always playing a instead of what you actually played. Positive regret means a was underused.

The goal is *no-regret*: a rule making $\max_a R^T(a)/T \rightarrow 0$. That is a weak-looking guarantee with a strong consequence, as we will see.

3.2 Regret matching

Definition 3.2 (Regret matching). Play each action with probability proportional to its positive regret so far:

$$\sigma^{T+1}(a) = \frac{[R^T(a)]^+}{\sum_{a'} [R^T(a')]^+}, \quad [x]^+ = \max(x, 0),$$

falling back to the uniform distribution when every regret is non-positive.

The intuition is exactly what it looks like: play what you wish you had played, in proportion to how much you wish it.

Example 3.3 (Worked: the keep-secret pennies). Take the class in which a seat holds one certain red and one certain black — statistic $(2, 1, 1)$ — and owes a discard. Its hand is public knowledge; its only private decision is *which card to keep*. The opponent simultaneously decides which color to play for. Keeping the card the opponent guesses loses; keeping the other wins. Payoffs are matching pennies:

	opp says R	opp says B
keep R	−1	+1
keep B	+1	−1

Run regret matching from zero regret. Iteration 1 is uniform, so each side mixes $(\frac{1}{2}, \frac{1}{2})$ and every action’s realized value equals the mean; regrets stay near zero and the iterates oscillate around $(\frac{1}{2}, \frac{1}{2})$ without settling on a pure action. The *average* strategy converges to $(\frac{1}{2}, \frac{1}{2})$ and the game value to 0 — which is the right answer, and is what the solver must reproduce for this class.

The example also previews a practical fact: fully mixed equilibria are the slow case for these methods. The iterates never settle, so the regret-based convergence meter decays more slowly here than at classes with a pure or near-pure answer.

3.3 Why sequential games need something more

Regret matching handles one decision. A round of Red/Black is a sequence of decisions under hidden information, and two complications appear.

First, a player does not know which *node* of the game tree they are at — only which *information set* (Section 4). A strategy must therefore be a single distribution per information set, used at every node in it.

Second, and less obviously, you cannot just run regret matching at each information set on its raw payoffs. The value of an information set depends on how often play actually arrives there, which depends on the strategies above it, including your own. Regret computed on raw payoffs mixes together “this action is bad” with “I rarely get here,” and the resulting quantity does not decompose into something you can minimize locally.

The fix is to strip out your own contribution to arriving.

3.4 Reach probabilities and counterfactual value

Definition 3.4 (Reach decomposition). For a node h , the probability of reaching it factors into three independent parts:

$$\pi(h) = \pi_{\text{own}}(h) \cdot \pi_{\text{opp}}(h) \cdot \pi_{\text{chance}}(h),$$

the contributions of the player’s own choices, the opponent’s choices, and chance. Write $\pi_{-i}(h) = \pi_{\text{opp}}(h)\pi_{\text{chance}}(h)$ for everything *except* player i .

Definition 3.5 (Counterfactual value). The counterfactual value of information set I for player i is

$$v_i(I) = \sum_{h \in I} \pi_{-i}(h) u_i(h),$$

the expected payoff weighted by *opponent-and-chance* reach only, as if player i had played to reach I with probability one.

Dropping π_{own} is the whole trick. It makes each information set’s regret independent of how the player behaves elsewhere, so the regrets at different information sets can be minimized separately and still add up to a bound on the regret of the whole strategy.

Definition 3.6 (Counterfactual regret; CFR). The counterfactual regret of action a at I is $R^T(I, a) = \sum_t (v_i^t(I, a) - v_i^t(I))$, where $v_i^t(I, a)$ is the counterfactual value of playing a and then continuing with the current strategy. *CFR* is: at every information set, choose the next strategy by regret matching on $R^T(I, \cdot)$.

3.5 CFR+

CFR+ modifies CFR in three ways. All three are practical accelerations, not changes to what is being computed.

Regret clipping. Cumulative regrets are floored at zero after every update rather than allowed to go arbitrarily negative: $R^t \leftarrow [R^{t-1} + \Delta]^+$. An action whose regret has gone deeply negative would otherwise take many iterations to become playable again after conditions change.

Linear averaging. The output of the algorithm is the *average* strategy, not the current one. CFR+ weights iteration t by t rather than uniformly, so late iterations — computed from better regrets — dominate.

Alternating updates. Rather than updating both players against a common snapshot, update one player at a time against the other’s current strategy.

Remark 3.7 (The average is the answer). This is the single most common source of confusion. The *current* strategy of a CFR run does not converge — Example 3.3 shows it oscillating forever. The object that converges to equilibrium is the running average. Every value, every meter, and every exported policy in this work refers to the average strategy.

3.6 What convergence does and does not guarantee

Regret matching gives $\max_a R^T(a) = O(\sqrt{T})$, so average regret vanishes as $O(1/\sqrt{T})$. CFR’s regret-decomposition theorem bounds a player’s total regret by the sum of counterfactual regrets over their information sets, so minimizing locally minimizes globally. And in a two-player zero-sum game, if both players’ average regrets are at most ε , the pair of average strategies is a 2ε -equilibrium.

Three caveats matter for what follows, and each is a place where a reader can be misled.

1. **The guarantee is about average regret, not about the value being right.** A profile can have small regret and still sit at a different point of a large equilibrium set. Where the equilibrium is not unique — and it is not, for the type-conditional values this paper bootstraps on — two runs can converge to different, equally valid answers.
2. **The guarantee is for the game you actually ran it on.** We do not run CFR+ on Red/Black; we run it on a decomposition of Red/Black (Section 5). Small regret in the decomposed game bounds exploitability *in the decomposed game*. Transferring that claim to the real game requires a separate argument or, better, a separate measurement.

3. **The bound requires perfect recall.** CFR’s decomposition theorem assumes players never forget their own past observations or actions. Within a round subgame this holds. Across the boundary between subgames it does not, by construction: the successor is entered through a public class that has forgotten the bid history and the path by which the bounds arrived.

Sections 7 and 8 exist because of these three caveats.

4 The Information Structure of a Round

This section is the specification. It defines, in order: what a player knows, what their private type is, what their private state is, what the decision nodes are, and finally what a *row* is.

4.1 What a player knows

Definition 4.1 (Information set). Fix a class (ℓ, s_0, s_1) and a seat i . An information set of seat i inside that class is a pair

$$I = (\text{public node, seat } i\text{'s private state}),$$

where the *public node* is everything both players can see at that moment — the class itself, the bid chain so far, and the flip counters with the colors revealed this contest — and the *private state* is everything seat i knows that its opponent does not.

Two seats at the same public node with different private states are at different information sets. Two *deals* that give seat i the same private state at the same public node are the *same* information set: that is precisely what seat i cannot distinguish.

4.2 Private type: (r, x) , and why x exists

Definition 4.2 (Private type). Seat i ’s type at a boundary is the pair

$$(r, x) = (\text{reds in its pre-discard hand, reds among the cards it has itself discarded}).$$

The public bounds truncate the first coordinate to $r_{\text{lb}}^i \leq r \leq n_i - b_{\text{lb}}^i$; the second is invisible to the opponent.

The first coordinate is unsurprising. The second deserves explanation, because it is easy to mistake for bookkeeping.

Discarded cards are out of play and can never be flipped, so x cannot affect any payoff directly. It is nonetheless *payoff-relevant information*, because the deck is finite. There are exactly 26 red and 26 black cards. A seat that has seen many reds among its own cards — in hand or discarded — knows the unseen remainder is red-poor, and can price its opponent’s hand accordingly. Concretely, a player who has seen k reds among their own five dealt cards should assign an unseen card probability $(26 - k)/47$ of being red, which ranges from 0.447 to 0.553 as k runs from 5 to 0.

So x is private information about the *deck*, not about the hand. Under an idealized infinite deck it would carry no information and could be dropped, collapsing the type space to r alone. It is retained because the real game is played with 52 cards.

4.3 Private states: what a seat actually chooses

A type is a piece of knowledge; a *private state* is a concrete realization the seat acts from. The two differ because a seat gets to choose its hand order, and the pending loser additionally chooses what to drop.

Definition 4.3 (Private state). For a seat that is **not** the pending discarder, a private state is

$$(\text{an ordering of its } (r, n_i - r) \text{ color multiset, } x),$$

one per admissible ordering, of which there are $\binom{n_i}{r}$.

For the **pending discarder**, the drop and the reorder are a single joint private choice, so a private state is

$$(\text{drop color} \times \text{ordering of the surviving } n_i - 1 \text{ cards, } x'),$$

where $x' = x + 1$ if the dropped card was red and $x' = x$ otherwise. Two paths that reach the same post-discard hand — dropping a red from r reds versus dropping a black from $r - 1$ — remain *distinct* states, because they imply different x' and therefore different beliefs about the deck.

Remark 4.4 (Why the drop and the reorder are joint). Separating them would let the reorder condition on a drop the opponent never sees, which is exactly the information leak the boundary placement exists to prevent. Treating them as one choice keeps the keep-secret decision and every response to it inside a single exactly-solved game.

Example 4.5 (Counting private states). Take the class $\ell = 0$, $s_0 = (2, 0, 0)$, $s_1 = (1, 0, 0)$ — seat 0 owes a discard from two cards, nothing is publicly certain about either hand.

Seat 0, the discarder, has $n_0 = 2$ so it has made $H - n_0 = 3$ discards, giving $x \in \{0, 1, 2, 3\}$; with no bounds, $r \in \{0, 1, 2\}$. That is $3 \times 4 = 12$ types. Its joint (drop, ordering) choices per r are: $r = 0$ must drop black, leaving B — 1 choice; $r = 1$ may drop either, leaving B or R — 2; $r = 2$ must drop red, leaving R — 1. Hence

$$\text{states}_0 = (1 + 2 + 1) \times 4 = \mathbf{16}.$$

Seat 1 has $n_1 = 1$, so 4 discards and $x \in \{0, \dots, 4\}$, and $r \in \{0, 1\}$: $2 \times 5 = 10$ types, each with exactly one ordering of a single card. Hence $\text{states}_1 = \mathbf{10}$.

4.4 Decision nodes

Within a round, the public structure is the round automaton of Paper I, built on the *in-round* sizes (n'_0, n'_1) . Two kinds of node carry decisions:

Bid nodes (color, chain), plus the pre-bid node. There are $2^{T'+1} - 1$ of them. The actor alternates with chain length.

Procurement nodes (color, chain, x, y) where x, y count cards already flipped off the bidder's and challenger's decks. Only the bidder acts. By Paper I's order-irrelevance lemma these form a lattice on the two counters, not a tree of flip sequences.

Resolutions are terminal within the round; they do not carry decisions, they carry values (Section 5).

4.5 Rows

Definition 4.6 (Row). A *row* is one (decision node, private state) pair belonging to the seat that acts at that node. Each row holds a cumulative-regret vector and a cumulative-strategy vector, one entry per legal action. The row count is the size of the regret table and the primary determinant of both memory and time.

The count is *not* a simple product of nodes and states, for two reasons: different nodes are acted on by different seats, and procurement nodes are masked by consistency — a bidder whose flipped prefix would not have matched the bid color cannot be at that node at all, so that row does not exist.

Example 4.7 (Rows, continued from Example 4.5). Same class. The starter is $1 - \ell = 1$, and the in-round sizes are $(1, 1)$, so the round graph has 7 bid nodes and 14 procurement nodes.

Of the 7 bid nodes, seat 1 acts at 3 and seat 0 at 4:

$$\text{rows}_{\text{bid}} = 3 \times 10 + 4 \times 16 = 94.$$

Of the 14 procurement nodes, seat 1 is the bidder at 8 and seat 0 at 6, which would give $8 \times 10 + 6 \times 16 = 176$ rows unmasked. The consistency mask removes 26, leaving $\text{rows}_{\text{proc}} = 150$. In total

$$\text{rows} = 94 + 150 = \mathbf{244}.$$

Table 1 gives the totals by layer. They are structural — derivable without solving anything — and they are what a run must budget for.

Remark 4.8 (A lossless reduction the present solver does not take). Those counts enumerate hand orderings in full, and they need not. Paper I shows that an ordering’s observable content is exactly its *leading run*: a hand opening with k cards of color a can never reveal anything at position $k + 2$ or beyond, because procurement halts at that seat’s first off-color card under either bid color, flipping removes no cards, and the next round reorders anyway. So orderings agreeing on (a, k) are indistinguishable in every continuation, and the $\binom{r+b}{r}$ orderings of an (r, b) multiset collapse to $r + b$ classes (one, if the hand is monochrome).

This is *lossless* — unlike the belief abstraction of Section 7, nothing is approximated — but the implementation described here does not exploit it: `build_privates` enumerates raw orderings. Taking it would remove about 12% of private states outright and, because the saving grows with depth (0% at $T' \leq 4$, 15% at $T' = 8$, 24% at $T' = 9$, 31% at the root) while the deep layers hold most of the rows, roughly **19%** of the total row count in Table 1 — some 62M of 336M, with $T' = 9$ alone contributing 40M.

Two caveats on the size. It requires $n \geq 4$ to appear at all, since $\binom{n}{r} = r + b$ for $n \leq 3$; and it is capped at 50% per composition, so lopsided hands ($r \in \{0, 1, n - 1, n\}$) save nothing. It is therefore not transformative alone. Its real value is that it partially funds a refinement that *costs* space: if a future version enriches the public statistic to make concealment visible (Section 7), this claws back a fifth of the bill.

Remark 4.9 (The root is the biggest game and one of the cheapest classes). The root is the full 5-versus-5 game. Its *public* structure is the largest anywhere — 2,047 bid nodes and 65,790 procurement nodes, more than any other class — and yet it holds only 815,456 rows, about four average layer-9 classes.

Table 1: Regret-table size by layer. “Classes” counts those with a real subgame; terminal classes hold a value and no rows. The distribution is extremely top-heavy: layers 8 and 9 hold 81.5% of all rows. The root ($T' = 10$) is the whole game and yet costs less than a percent — see Remark 4.9.

Layer T'	Classes	Rows	Mean rows/class	Share
2	36	5,424	150	0.002%
3	132	70,056	530	0.021%
4	330	546,656	1,656	0.163%
5	686	3,181,392	4,637	0.948%
6	1,104	13,760,056	12,463	4.100%
7	1,290	43,581,784	33,784	12.985%
8	1,260	106,941,312	84,874	31.862%
9	882	166,740,672	189,048	49.678%
10 (root)	1	815,456	815,456	0.243%
all	5,721	335,642,808		

The reason is that rows are the product of public nodes and *private states*, and at the root the private side is at its smallest. Nobody has discarded yet, so $x = 0$ for both seats and the type space collapses from (r, x) to r alone: 6 types per seat, and $\sum_r \binom{5}{r} = 2^5 = 32$ private states. Contrast a layer-9 class, where each seat has made discards, x ranges over several values, and the private state count is multiplied accordingly.

This inverts the naive intuition that the game gets cheaper as it gets smaller. It does not: the public game shrinks with T' , but the private state space *grows* with the number of discards, and the second effect dominates. The expensive layers are the middle-to-late ones, where hands are still large enough for many orderings and enough cards have been discarded for x to range widely.

4.6 Entry beliefs

A subgame cannot be solved without a distribution over the type pairs it starts from. We use the *bounded hypergeometric* law: treat both hands as drawn from the residual 26/26 budget and renormalize over the region the public bounds allow.

This is the one modeling choice in the design, and Section 7 is about its consequences. Note what it is not: it is not the exact posterior, because the exact posterior depends on how the opponent has been playing.

5 Decomposition

5.1 Subgames

Definition 5.1 (Round subgame). The subgame rooted at a non-terminal class (ℓ, s_0, s_1) consists of: the pending discarder’s joint (drop, reorder) choice; the other seat’s reorder; the bidding phase; and procurement, ending at the resolution *before* the next discard.

Two boundary cases. *Terminal* classes — those whose pending discarder holds a single card

— have no subgame at all: the forced discard eliminates that seat and the value is ± 1 . The *root* is the opposite special case: $\ell = -1$, so no discard is pending, the pre-stage is just the two reorders, and the bidding starter is seat 0 by convention rather than $1 - \ell$. Everything downstream of the pre-stage is identical.

5.2 Backward induction over layers

Every resolution of a subgame at layer T' identifies a successor class at layer $T' - 1$: apply the realized discard's decay to the entry bounds, raise them to the colors seen this round, and record the new loser. Because T' strictly decreases, the classes form a layered DAG and can be solved bottom-up.

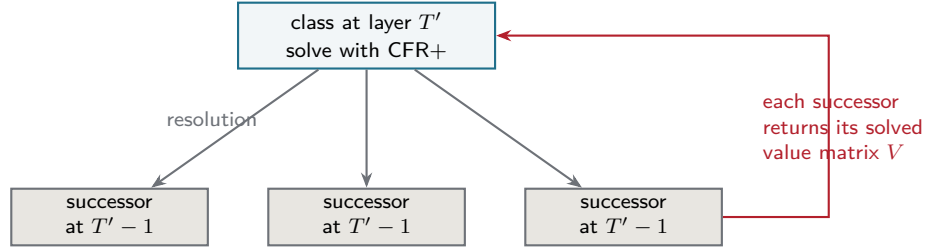


Figure 1: One step of the recursion. A class is solved with CFR+ over its own round; its terminal payoffs are type-conditional values returned by already-solved successors one layer down.

5.3 Bootstrapping

Definition 5.2 (Type-conditional value matrix). Write m_i for the number of seat i 's types. The solution of a class returns a matrix $V \in \mathbb{R}^{m_0 \times m_1}$, where $V[t_0, t_1]$ is the seat-0 value when the seats hold types t_0, t_1 and both play the solved average strategy.

A parent's resolution looks up its successor class and reads off the entry of V matching the types the seats hold at that moment. Only V is needed to solve a parent, which is why high layers can be solved against a values-only store rather than full child solutions.

Remark 5.3 (V is not unique). Different equilibria of the same subgame can produce different V while agreeing on the aggregate value. Where a private coordinate is payoff-irrelevant but belief-relevant — x is exactly this — the solver may split a mixed strategy into near-pure branches indexed by that coordinate, and the individual entries of V are then arbitrary across the equilibrium set. The aggregate is pinned; the cells are not. This is a measurement trap, and Section 8 returns to it.

6 Vector-Form CFR+

6.1 Why not walk information sets

The textbook presentation of CFR walks the game tree recursively, visiting one node at a time. That is a poor fit here. The number of private states per seat is large and their structure is identical — every private state of a seat faces the same public node with the same legal actions. Walking them individually means millions of tiny scalar operations.

Instead we hold, at each public node, a *matrix* whose rows are the acting seat’s private states and whose columns are the legal actions. An iteration then becomes a sequence of dense matrix operations over public nodes, with the private dimension vectorized.

6.2 Layout

Per class, for each seat i :

Object	Shape	Meaning
<code>states</code>	P_i	private states (Definition 4.3)
<code>type_idx</code>	P_i	which type each private state belongs to
<code>regret[node]</code>	$P_i \times A_{\text{node}}$	cumulative regret, one row per state
<code>strat_sum[node]</code>	$P_i \times A_{\text{node}}$	cumulative strategy, linearly weighted
<code>reach</code>	P_i	current reach for each private state
<code>entry_wt</code>	$ \text{types}_0 \times \text{types}_1 $	bounded-hypergeometric prior

The row count of Definition 4.6 is exactly $\sum_{\text{nodes}} P_{\text{actor}(\text{node})}$, restricted by the consistency masks — which is what Example 4.7 computed by hand.

6.3 Consistency masks

A private state is not compatible with every public node. If a procurement node records that three cards came off a seat’s deck and all matched the bid color, then any private state whose top three cards are not all that color is impossible there. Rather than branch, we precompute a boolean mask per (node, seat) and zero the corresponding rows. Masked rows are excluded from the row count because they can never be reached.

6.4 One sweep

An iteration, for the player being updated:

1. **Strategy.** At each node, regret-match the regret matrix rowwise to get a strategy matrix, clipping negatives (CFR+).
2. **Forward.** Propagate reach from the entry weights through the pre-stage choice and then down the public graph, multiplying by the acting seat’s strategy and applying masks.
3. **Backward.** Compute counterfactual values at resolutions by contracting the successor’s value matrix V against the opponent’s reach, then propagate upward, accumulating per-action values.
4. **Regret.** Add (action value — node value), weighted by opponent-and-chance reach, into the regret matrix; clip at zero.
5. **Average.** Accumulate $t \cdot \pi_{\text{own}} \cdot \sigma$ into the strategy sum.

Then swap players. The average strategy is recovered by normalizing `strat_sum` rowwise.

Remark 6.1 (Averaging must not be gated by opponent reach). The accumulation in step 5 must run at every information set the updating player can reach on their own, regardless of whether the opponent currently reaches it. Gating on opponent reach freezes rows the opponent has temporarily stopped visiting, leaving early garbage in the average and making the result spuriously exploitable.

6.5 Cost model

Time per class scales as

$$\text{cost} \approx \frac{\text{rows} \times \text{iterations}}{\text{throughput}},$$

with throughput in row-iterations per second. Rows are structural (Table 1); iterations are what convergence demands and are reported in Section 9. The important structural fact is the top-heaviness: half the total row count lives at layer 9 alone, and layers 8 and 9 together are 81.5% of the work. Any scheduling decision should be made about those two layers first.

6.6 The second-generation solver

The first full solve (Section 9.1) was produced by the machinery above, run at $\text{tol} = 10^{-3}$ on a cloud fleet. Its autopsy (Section 9.2) motivated three changes that together turned a \$23, twelve-hour fleet job into a two-hour laptop job at $10\times$ – $100\times$ tighter tolerance. Each is a structural observation about the game, not an optimization trick, which is why they belong in this paper rather than in release notes.

Ordering collapse. A seat’s private state included its chosen deck ordering. But orderings that present the same color *prefix behavior* are strategically identical — the opponent can only ever observe a prefix — so ordering-sensitive states quotient into *order classes*, exactly as Paper I’s Lemma on order irrelevance collapses procurement states. This is the same state-versus-information distinction doing work a second time: the information a player can ever transmit through an ordering is its prefix structure, and solving over anything finer is solving redundant copies. The collapse shrinks private state spaces by an order of magnitude at the deep layers.

Seat-mirror reflection. A class and its seat-swapped image are the same subgame with the labels exchanged, so only one of each pair is solved and the other is produced by reflection. This makes mirror antisymmetry — the first solve’s sharpest measured failure — hold *exactly, by construction*, and it halves the work. The honest check then moves one level up: re-solving a sample of reflected classes directly and comparing prior-weighted aggregates gives residuals of 1.5×10^{-6} to 8.9×10^{-6} , against first-solve mirror gaps of up to 0.114.

Remark 6.2 (Equilibrium multiplicity is not error). Cell-wise, a re-solved class and its reflection can disagree by as much as 6.7×10^{-2} in individual type-pair conditional values while agreeing to 10^{-6} in every prior-weighted aggregate. Zero-sum equilibria pin values, not solutions; conditional-value cells are selection-dependent. Consumers must never compare cells across independently produced solutions — a rule that returns with force in Paper III, where per-type promises are exactly such cells.

The kernel. Profiling showed the solve was interpreter-bound: tiny dense matrices under huge node counts. Compiling the sweep (and, later, the best-response sweeps of Section 8) into a JIT kernel recovered the arithmetic throughput the cost model above assumes. With all three changes the full game solves locally in 2h02m at $\text{tol} = 10^{-5}..10^{-4}$, and the pipeline is deterministic: re-running it reproduces the solve bit-for-bit.

7 What Is Approximated

The algorithm above solves each subgame essentially exactly. The approximation lives entirely in how subgames are joined.

7.1 The belief reset

To solve a subgame we need a distribution over its starting type pairs. We re-derive it from the public class using the bounded-hypergeometric law of Section 4.6. We do *not* carry down the posterior that the actual history would have induced.

Paper I makes precise what that discards. Its support-completeness theorem (Thm. 11.1 there) shows the class determines the support of the public posterior exactly — the certainty bounds are sound and tight. But the shape of the distribution on that support is not determined: two histories reaching the same class can leave an observer with materially different beliefs, because the bound-folding rule remembers a bound’s height while forgetting whether it is fresh evidence or the decayed residue of a taller one. Cross-round bid history is discarded outright.

7.2 Consequences

1. **The solved profile is an equilibrium of a different game.** Call it G' : Red/Black with beliefs reset to the class prior at every boundary. CFR+’s guarantees apply to G' . In the real game the profile is a candidate strategy whose exploitability must be measured.
2. **Round-local convergence cannot certify the composition.** A class can be solved to arbitrarily small round-local exploitability while the values it inherited from below are drawn from a different point of a non-unique equilibrium set (Remark 5.3).
3. **The error is not automatically small.** Appendix A documents a boundary placement that produced a 9.5 percentage point error on a concrete position while remaining internally consistent by every round-local measure. That case motivated the present boundary; it is not evidence that the current one is exact.

None of this is an argument against decomposing. It is an argument for measuring, which is the next section.

8 Measurement

Three instruments, three different claims. Confusing them is the main way to overstate a result.

8.1 Round-local exploitability

Definition 8.1 ($\varepsilon_{\text{round}}$). For a single class, hold the solved continuation values fixed and compute both players’ exact best responses against the average profile *within that round*. The sum of their gains is the round-local exploitability.

Certifies: that CFR+ has converged on this subgame, given its inherited values. **Does not certify:** anything about composition. It is computed with the successor values held fixed, so it is blind by construction to whether those values were right.

8.2 Composed exploitability

Definition 8.2 ($\varepsilon_{\text{comp}}$). Best-respond across the whole class DAG — in the round and in every successor round — against the cached average profile.

Certifies: exploitability in G' , the belief-reset game. **Does not certify:** exploitability in Red/Black. Two sub-meters are worth distinguishing: a *lower bound*, obtained because any concrete deviating strategy’s gain bounds the true best response from below; and an *exact* figure for a redeal variant in which types are re-drawn at every boundary, where scalar continuations make backward induction exact. Neither removes the belief reset — they measure inside it.

Incident — the negative exploitability (2026-07)

An early composed-exploitability run reported $\varepsilon_{\text{comp}} < 0$, which is impossible for a true best response against any profile. The cause was the lower-bound structure itself: child-level best response maximizes a child-posterior-weighted total, so a parent can weight exactly the type pairs sitting *below* the profile’s value, and the composition of locally-best responses is not a globally best response. The number was a valid lower bound doing exactly what lower bounds do. Lesson: label every meter with its bound direction *in the code*, and treat any “impossible” reading as a definition surfacing, not a bug to patch away. A second incident lived one layer down: the best-response context loaded per-state tables into per-shared-row storage, silently misaligning every class where the discarder collapse makes those indexings differ — voiding every composed figure computed before the fix. Meters need gates too: the kernel best response is validated against an independent reference implementation on 1,184 classes to $|\Delta| \leq 4.4 \times 10^{-16}$ before any number it produces is reported.

8.3 The perfect-recall oracle

Definition 8.3 (Oracle bracket). Build the explicit perfect-recall game tree over the raw engine, with information sets indexed by a player’s full own history and every public observation, so nothing the real game remembers is forgotten. Then bracket the value with exact best responses.

Certifies: the true value, in the real game, with no belief reset anywhere. **Limitation:** the tree is built explicitly, so this is restricted to small classes. Until the next subsection it was the only instrument here that speaks about Red/Black rather than about G' , and it is the one that scales worst — a combination worth stating plainly rather than burying.

8.4 Instruments for the real game

The first solve was evaluated with the three instruments above and the seat mirror. The distance results of Section 9.6 required two more, both of which certify statements about G itself and both of which scale.

Definition 8.4 (TV meter). Play the profile against itself; at every boundary compute the exact public posterior μ_h over type pairs (a forward pass over the factored belief family of Paper I, validated against brute-force replay to 10^{-12}) and report the reach-weighted total variation $\text{TV}(\mu_h, \lambda_\kappa)$ against the class prior the solver used instead.

Certifies: how much belief the reset actually destroys, on the profile’s own play path — the size of the modeling choice, measured where it is being made.

Definition 8.5 (LBR probe). A local best responder plays one seat against the stored profile. It maintains the exact opponent-type posterior conditioned on the public history and its own private trajectory, prices each of its actions by contracting the profile’s per-class continuation values against that posterior, and plays the argmax. Its measured win-rate gain over the profile-vs-profile baseline (mirrored seeds, Wilson interval) is a *certified lower bound* on the profile’s real-game exploitability ε_G .

Certifies: $\varepsilon_G \geq$ the measured gain — certification comes from the games actually won, not from the derivation, so it holds no matter how crude the responder is. **Does not certify:** any upper bound. A profile the probe cannot beat is not thereby safe; upper bounds are Paper III’s whole subject.

8.5 Summary

Table 2: What each instrument certifies. No single row is sufficient, and the rightmost column is the one that matters for any claim about the real game.

Instrument	Scope	Certifies about	Scales?
$\varepsilon_{\text{round}}$	one class	convergence, given inherited values	yes
$\varepsilon_{\text{comp}}$	class DAG	exploitability in G' (lower bd.)	yes
Oracle bracket	one small class	the true value in Red/Black	no
Seat mirror	class pair	internal consistency	yes
TV meter	play path	size of the belief reset in G	yes
LBR probe	whole game	ε_G lower bound, certified by games	yes

The seat mirror deserves a note. A class and its seat-swapped image must have exactly opposite values, for any correct solver. At layers where a subgame is the entire remaining game the mirror is tight; once classes bootstrap from other classes, mirror discrepancy becomes a cheap proxy for composition error — precisely the quantity $\varepsilon_{\text{round}}$ cannot see. It is the only instrument in the table that is both scalable and sensitive to composition, which makes it the practical tripwire for a large run.

8.6 The defensible claim, and the two-axis rule

Not “the game is solved.” The defensible form is a conjunction: a stated round-local tolerance across all classes; a composed exploitability in G' ; an oracle agreement on the small classes where it can be computed; mirror consistency as a function of depth; and an explicit statement that the profile is an equilibrium of G' rather than of Red/Black. Section 9 reports all of these — and then reports the two real-game instruments alongside, because the central empirical finding of this paper is that **the two axes decouple**: profile variants that differ by 0.05 per game in G' -exploitability measure identically on every real-game instrument, and the profile whose G' meters read 10^{-4} is exploitable for ≥ 0.598 per game in G . Any report that cites one axis without the other is, on this evidence, misleading regardless of which axis it picks.

9 Results

The abstraction G' has been solved twice: all 5,721 classes, every layer from $T' = 2$ to the root, first at $\text{tol} = 10^{-3}$ on a cloud fleet, then — after the first solve’s autopsy — at $\text{tol} = 10^{-5}..10^{-4}$ on a laptop with the second-generation solver of Section 6.6. This section reports both, in chronological order, because the first solve’s failure is what makes the second one’s claims legible; then the equilibrium-selection experiments; then the measured distance from G' to G (Section 9.6), which is the half of this section we expect to age best.

9.1 The first solve

Table 3: Convergence by layer at $\text{tol} = 10^{-3}$, $\text{max_iters} = 6000$. Every class in every layer converged; no class hit the cap.

Layer T'	Classes	Median iters	Max iters	Max $\varepsilon_{\text{round}}$	Rows
2	36	275	3,075	9.9×10^{-4}	5,424
3	132	200	2,075	9.9×10^{-4}	70,056
4	330	125	1,850	1.0×10^{-3}	546,656
5	686	100	1,575	1.0×10^{-3}	3,181,392
6	1,104	150	1,475	1.0×10^{-3}	13,760,056
7	1,290	200	1,025	1.0×10^{-3}	43,581,784
8	1,260	250	900	1.0×10^{-3}	106,941,312
9	882	350	975	1.0×10^{-3}	166,740,672
10	1	475	475	9.8×10^{-4}	815,456
all	5,721				335,642,808

Layers 2 through 9 ran on a 28-physical-core cloud VM in 43,653 s (12.1 hours) for about \$23; the root ran locally in 2,019 s (33.6 minutes). The root is one class, so it cannot be parallelized and core *speed* is all that matters — an Apple M-series performance core measured $\approx 2.2\times$ a cloud core here, which is why bringing a single class home beat leaving it on the fleet.

Two practical findings. First, max_iters must be generous: fully mixed classes — the keep-secret matching-pennies shape of Example 3.3 — need $\sim 2,700$ iterations against a median of a few hundred, and the common default of 800 silently truncates about 17% of the shallowest layer, failing the ε gate for reasons that have nothing to do with the game. A high cap is nearly free because the solver exits at tolerance.

Second, median iterations are **U-shaped** in depth: 275 at $T' = 2$, falling to 100 at $T' = 5$, then climbing to 350 at $T' = 9$ and 475 at the root. Extrapolating a cost model from the shallow layers alone reads the descending arm and underestimates the deep ones.

The first solve priced the root at $v = +0.0266$ (51.33% to the opening seat) — a number the next subsection retires.

9.2 The autopsy: what the meters saw and did not

Every class converged: $\varepsilon_{\text{round}} \leq 10^{-3}$ everywhere, with no class hitting the iteration cap. On the round-local meter the solve is uniformly green.

That meter is not the story. Seat-mirror antisymmetry must hold *exactly* — $v(\ell=0, A, B) = -v(\ell=1, B, A)$ — and over all 5,720 mirror pairs it does not:

Layer T'	2	3	4	5	6	7	8	9
Worst gap	0.0002	0.0056	0.0141	0.0288	0.1142	0.0767	0.0495	0.1146

Median 0.0014; worst 0.1146, which is over $100\times$ that class’s own $\varepsilon_{\text{round}}$ and 5.7 percentage points of win rate. This is exactly the failure Section 8 predicted: $\varepsilon_{\text{round}}$ certifies a class against its inherited values and is blind by construction to whether those values were right.

Two features of that table deserve emphasis. The gap does *not* decay with depth — it is bimodal, peaking at $T' = 6$ and again at $T' = 9$. And $T' = 9$, the worst layer in the entire solve, is the layer every game enters on its second round. The best-calibrated numbers are the ones a player sees last.

9.3 The solve of record

The second-generation solver (Section 6.6) re-solved the entire game in 2h02m on a laptop, eight workers, at layer tolerances of 10^{-5} to 10^{-4} — $10\times$ to $100\times$ tighter than the first solve — with root $\varepsilon_{\text{round}} = 4.9 \times 10^{-5}$ and six classes above their layer tolerance (all within a 10^{-5} band of it). Mirror antisymmetry is exact by construction; the reflection residual against independently re-solved mirrors is 1.5×10^{-6} .. 8.9×10^{-6} . The pipeline is deterministic and reproduces itself bit-for-bit.

$$v(\text{root}) = +0.02449 \implies \mathbf{51.22\%} \text{ to the player who opens.}$$

This is the value of G' of record. The first solve’s 51.33% was carrying about 0.11 points of composition error — consistent with its own mirror-gap table, and a useful calibration of what $\text{tol} = 10^{-3}$ plus belief-reset bootstrapping actually delivers. The opening seat’s edge sits between the two small games solved by hand in Paper I (51.92% at $T' = 2$, 52.83% at the four-card root), and the oracle bracket agrees with the solved value to 1.4×10^{-4} on every class small enough to bracket ($T' \leq 3$).

9.4 Equilibrium selection, priced

Zero-sum equilibria pin values, not strategies, so within measured indifference there is freedom to select. Two selections were built and priced — and the price list is itself a finding:

Profile variant	G' root value	$\varepsilon_{\text{comp}}$ in G' (/game)	suicide rate (raw)
base (solve of record)	+0.02449	0.4533	3.6%
dominance-pruned	+0.02397	0.5053	0
concealment tie-break	+0.02449	0.5003	2.05%

The pruned variant removes the suicide flip (Section 9.6) from the action set; it surrenders 5.2×10^{-4} of G' root value and +0.052/game of G' -composed exploitability — the measured price of refusing an action that is optimal in G' and dominated in G . The tie-break selects, among value-equal actions, the one that reveals less; its purity costs +0.047/game of G' exploitability, roughly $50\times$ the naive per-decision band intuition, because the band bounds the value shift, not the composed-best-response shift. **Neither price is visible on any real-game instrument:**

LBR, raw and guarded arenas, and the bot battery all measure the three variants as identical. That is the two-axis decoupling of Section 8, measured.

9.5 Playing the profile

The solved profiles were mapped into the game engine and evaluated over mirrored pairs (each deal played from both seats; 2,000 pairs per cell; play-time dominance guards on):

Profile	vs gen6 CFR	vs heuristic	self-play seat-0
first solve (historical)	78.85% [77.6, 80.1]	87.10% [86.0, 88.1]	51.0%
solve of record	79.97% [78.7, 81.2]	87.55% [86.5, 88.5]	50.4%
dominance-pruned	81.55% [80.3, 82.7]	85.25% [84.1, 86.3]	51.7%
concealment tie-break	80.13% [78.9, 81.3]	86.88% [85.8, 87.9]	51.0%

Across every evaluation the profile drove *every* decision: zero fallbacks, zero out-of-range positions, zero on-demand solves. Since a misplaced boundary or a pre- versus in-round size confusion surfaces as a fallback, that is meaningful evidence the class mapping is correct, and every joint discard-and-reorder commitment replayed exactly. Self-play seat-0 rates bracket the solved root (51.22%) within their intervals: table, mapping and engine agree.

The deployed profile is the concealment tie-break variant: value-neutral against every measured opponent, strictly more concealing by construction, and it halves the raw suicide mass. The pruned variant trades 2.3 points against the heuristic for 1.6 against gen6 — a reminder that against non-equilibrium opponents, removing a G' -optimal action changes matchups in both directions.

Two caveats frame the table. The win rates do not measure distance from equilibrium: none of these opponents probes the abstraction. And the raw profile is not safe alone — with guards off, self-play logged suicide flips at 3.6% of procurement decisions and hundreds of impossible bids; the guard layer removes both, which also means **guards-on parity between profile variants is a floor, not a finding**: guards prune exactly the action classes where table improvements would show.

9.6 The distance to the real game

Everything above is internal to G' or to a bot ladder. This subsection reports what the real-game instruments say about the solve of record, and it is the paper’s destination.

The reset destroys most of the belief. The TV meter over 2,000 self-play games (11,320 boundaries): reach-weighted $\text{TV}(\mu_h, \lambda_\kappa) = \mathbf{0.842}$ [0.839, 0.845], monotone with depth — 0.66 at the first boundary, 0.95 by the endgame layers. The class prior at depth is nearly orthogonal to the belief an exact observer holds. A cross-fitted refinement ladder localizes the loss: replacing the prior with the *true class-conditional posterior mean* closes the gap only from 0.842 to 0.671, and enriching the class key with freshness bits (distinguishing Paper I’s Path A from Path B) adds only ≈ 0.024 more. Most of the destroyed belief is *within-class* variation that no small public statistic reaches — the tilts of Paper I’s factorization, which only actual belief tracking recovers.

The abstraction’s equilibrium is qualitatively wrong in G . The sharpest single result:

Theorem 9.1 (The suicide flip is G' -optimal and G -dominated). *Call a suicide flip a bidder’s choice to flip its own deck when its own top card is certainly off-color. In real Red/Black the action is weakly dominated: flipping the challenger’s deck preserves the winning branch, loses the same contests, and reveals the opponent’s card instead of one’s own. Yet in G' , at a probe class solved with successor values held fixed, the exact equilibrium plays the suicide flip on path with probability 0.97; exact best response prices it +0.026 over the dominating alternative; and a solve with the action masked cannot converge under an unmasked best response (its exploitability floors at 0.115).*

The mechanism is the belief reset laundering the self-reveal: busting your own known card raises your public bound, and the successor re-derives beliefs from the class — a diffuse prior — where a real opponent’s posterior after the bid-then-deliberate-bust line would be devastating. The 3.6% on-path suicide rate of the raw profile is not unconverged residue; it is *correct play in the wrong game*. This also retires the first solve’s reading of the same phenomenon (the “noise laundered into confident advice” hypothesis): the second solve converged two orders tighter and the rate did not move. Convergence cannot fix an abstraction artifact at any tolerance; only the action set or the beliefs can.

Certified real-game exploitability. The LBR probe (Definition 8.5), 3,000 mirrored games per profile, zero fallbacks:

$$\varepsilon_G \geq 0.598 [0.569, 0.626] \text{ per game (value range } [-1, 1]),$$

identically for the solve of record and for *every* selection variant — against G' -composed exploitability differences of 0.05/game between those same variants, and per-round convergence of 10^{-5} . The two axes decouple completely, in both directions: the G' meter prices exploitability in exactly the abstraction whose belief structure the real game does not share.

Incident — head-to-head is not a solve meter (2026-08)

The second solve, converged 100× tighter to G' than the first, *lost* their raw head-to-head arena: 48.2% [46.8, 49.6], $p = 0.014$, over 4,800 games — a result that would have justified reverting it. An overnight campaign of 216,000 games proved the matchup a dead tie (50.04% [49.8, 50.3]), and deterministic replay reproduced the 48.2% on its exact seed family: an unlucky draw amplified by pair-correlated p -values. The equilibrium guarantee floors value inside G' only; in G neither profile is an equilibrium, and two non-equilibria are not ordered by their distance to the G' solve. Lesson: a gate of the form “new solve beats old solve in games” tests pairwise interaction plus sampling noise, not solution quality. The honest instruments are the composed meters and the probe.

9.7 What is established, and what is not

Established: G' is solved, exactly, reproducibly, with every meter green and the oracle in agreement where it can see; and the distance from G' to G is measured — TV 0.842, $\varepsilon_G \geq 0.598$, one on-path dominated action with its mechanism identified.

Not established: any upper bound on ε_G , for this or any profile. Every real-game instrument in this paper bounds from below. The evidence that the gap is closable is already in hand — re-solving boundaries under the exact posterior (the tilts this paper discards) cuts the probe’s

certified bound from 0.598 to 0.363 at a shallow depth gate, 0.183 deeper, and ≈ 0.013 with full carrying, a complete dose-response — but carrying beliefs soundly, and *certifying* the result against all opponents rather than one probe family, is a different problem with different mathematics. That problem, its theorem, its instruments, and its current honest status are Paper III.

A An Earlier Boundary, and Why It Failed

The boundary of Definition 5.1 places the pending discard at the *start* of a subgame. An earlier design placed it at the *end* — subgames ran *reorder* $\rightarrow$ *bid* $\rightarrow$ *procure* $\rightarrow$ *discard*, matching the rules’ own phrasing. That choice is more natural to write down and it is wrong, in a way worth recording because the failure was invisible to every round-local check.

The counterexample. Consider a seat holding one card, known to be red, first to act, against a seat holding a known red and a known black. By a two-stage matching-pennies argument the true value of this position is exactly 25% to the first seat. The end-of-round decomposition priced it at 15.5%.

The mechanism. With the discard at the end of a subgame, the loser’s keep-secret decision falls in one subgame while every response to it falls in the next. The successor was solved standalone against a class prior, which committed it to near-pure responses conditioned on the opponent’s type. The parent’s discard could then choose the card that beats that already-frozen book — an exploit the real game forbids, because the response cannot be conditioned on a secret it has not seen. The keep-secret and the response are simultaneous in the real game, and the boundary had severed them.

Why it was invisible. Round-local exploitability was small. Composed best-response brackets confirmed the 15.5% tightly — but *within the belief-reset game*, which is what Section 8 warns about. The number was internally consistent and externally wrong. Only an oracle that refused the belief reset could see it.

The fix. Move the boundary one discard earlier, so that the keep-secret decision and every response to it live inside a single exactly solved subgame. That is Definition 5.1, and it is why the pending discarder’s drop and reorder are a joint private choice (Definition 4.3).

The lesson, stated generally. A decomposition boundary must never cut a coupling between moves that the real game makes at the same time. Where it does, each side gets solved against a frozen model of the other, and the error can be large while every meter computed inside the decomposition reports success.

B Notation

Continuous with Paper I; new symbols marked †.

Symbol	Meaning
H	hand size at the deal; $H = 5$
n_i, T	hand size / cards in play at a boundary (<i>pre</i> -discard)
n'_i, T'	hand size / cards in play during the round; T' is the layer index
ℓ	seat owing a discard; -1 at the root
$s_i = (n_i, r_{\text{lb}}^i, b_{\text{lb}}^i)$	seat i 's public statistic
(r, x)	private type: reds held, reds among own discards
$P_i \dagger$	number of seat i 's private states
$I \dagger$	information set
$R^T(I, a) \dagger$	cumulative counterfactual regret
$\pi_{\text{own}}, \pi_{\text{opp}}, \pi_{-i} \dagger$	reach probabilities
$v_i(I) \dagger$	counterfactual value
$V \dagger$	type-conditional value matrix of a solved class
$G' \dagger$	the belief-reset game the solver actually solves
$\varepsilon_{\text{round}}, \varepsilon_{\text{comp}} \dagger$	round-local / composed exploitability